

Task 12 — : Build an AWS Cost and Resource Review

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Date: 2026-09-05

1. Approach and Architecture

The objective of this task was to perform a basic cloud hygiene review on a fictional AWS environment containing multiple EC2 instances, storage resources, and IAM identities. The management team wanted to know what resources were being used, what was unnecessary, and where costs could be reduced.

My approach was to systematically inventory all AWS resources, identify unused or underutilized resources, review IAM permissions, estimate costs, and provide actionable recommendations. I used the AWS Console to gather information about EC2 instances, S3 buckets, and IAM users. I then analyzed the data to separate confirmed issues from assumptions, created a prioritized findings list, and prepared a management-ready cloud hygiene report.

2. Implementation Steps

Step 1 – Inventory the AWS Resources: I started by listing all resources in my AWS account. This included EC2 instances, S3 buckets, and IAM users. I documented each resource along with its current state and purpose.

Step 2 – Identify Resources That Appear Unused: I reviewed the resource list and identified resources that had no recent activity or were stopped. These included a stopped EC2 instance and old S3 buckets that were created for previous tasks.

Step 3 – Review EC2 Instance Configurations: I checked each EC2 instance's configuration, including its instance type, state, and purpose. I noted that most instances were t3.micro and were either running or stopped.

Step 4 – Review Storage Usage: I reviewed all S3 buckets and EBS volumes to understand storage usage and identify any unnecessary storage costs.

Step 5 – Review IAM Users and Permissions: I reviewed each IAM user and the policies attached to them to understand who had access to what.

Step 6 – Identify Potentially Excessive Permissions: I looked for users with broad permissions like AdministratorAccess or full S3 access and noted them as potential risks.

Step 7 – Identify Resources That May Generate Unnecessary Costs: I identified resources that were either stopped or unused but still incurring storage costs, such as stopped EC2 instances and old S3 buckets.

Step 8 – Separate Confirmed Issues from Assumptions: I distinguished between confirmed issues, such as stopped instances and unused users, from assumptions, such as whether certain buckets might still be needed.

Step 9 – Create a Prioritized Findings List: I created a list of findings prioritized by their impact on security and cost.

Step 10 – Recommend Appropriate Cleanup Actions: I recommended actions like terminating stopped instances, deleting unused buckets, and removing excessive permissions.

Step 11 – Explain Which Changes Require Additional Verification: I noted that deleting S3 buckets and removing permissions should be verified before implementation to avoid accidental data loss.

Step 12 – Create a Simple Resource-Management Checklist: I created a checklist for regular resource reviews to prevent future waste.

Step 13 – Estimate Potential Cost-Saving Opportunities: I estimated the cost savings from implementing the recommended cleanup actions.

Step 14 – Document Security and Operational Observations: I documented observations about missing security controls like MFA and CloudTrail.

Step 15 – Prepare a Management-Ready Cloud Hygiene Report: I compiled all findings and recommendations into a professional report suitable for management review.

3. Key AWS Concepts Learned

- **Cloud Hygiene:** I learned the importance of regularly reviewing AWS resources to identify waste, reduce costs, and improve security.
 - **Resource Inventory:** I understood the need to maintain a clear inventory of all resources, including their state, purpose, and cost.
 - **Unused Resource Identification:** I learned how to identify resources that are not being used, such as stopped EC2 instances and old S3 buckets.
 - **IAM Review:** I practiced reviewing IAM users and permissions to identify excessive or unnecessary access.
 - **Cost Estimation:** I learned how to estimate costs for different AWS services, including EC2, S3, and EBS.
 - **Prioritization:** I understood the importance of prioritizing findings based on their impact on security and cost.
 - **Cleanup Verification:** I learned that cleanup actions should be verified before implementation to avoid accidental data loss.
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4. Challenges Faced

- **Identifying Unused Resources:** The biggest challenge was identifying which resources were truly unused versus those that were being used for active tasks. For example, I had to check the last activity date for S3 buckets and review the purpose of each EC2 instance.

- **Estimating Costs:** Estimating the cost of resources was tricky because I didn't have access to detailed billing data. I had to rely on estimated rates and make reasonable assumptions.
 - **IAM Permission Review:** Reviewing IAM permissions was challenging because I had to understand the purpose of each user and determine whether their permissions were appropriate.
 - **Separating Assumptions from Confirmed Issues:** I had to carefully separate confirmed issues, such as stopped instances, from assumptions, such as whether certain buckets were still needed.
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5. Solutions Implemented

- **Resource Inventory:** I created a detailed inventory of all resources, including their state, purpose, and cost impact. This helped me identify unused resources more easily.
 - **Unused Resource Identification:** I used the AWS Console to check the last activity date for S3 buckets and reviewed the state of each EC2 instance. I also checked IAM users for any recent activity.
 - **Cost Estimation:** I used published AWS pricing data to estimate the cost of EC2 instances, S3 storage, and EBS volumes. I calculated the monthly cost based on current usage.
 - **Prioritized Findings:** I prioritized findings based on their impact on security and cost, with high-priority items being those that posed a security risk or generated significant costs.
 - **Verification Recommendations:** I recommended that any changes, such as deleting S3 buckets or removing permissions, should be verified before implementation to avoid accidental data loss.
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6. Security Considerations

- **Excessive Permissions:** I identified that the task3-iam-user had AdministratorAccess, which is a high-risk permission. This user should have only the permissions required for their role.
 - **Unused IAM Users:** I found that users like hr-admin-user, mytask, and operations-user had no recent activity and should be deleted or disabled.
 - **Missing MFA:** None of the IAM users had MFA enabled, which is a significant security gap. I recommended enabling MFA for all users.
 - **Missing CloudTrail:** CloudTrail was not enabled, which meant API calls were not being logged. I recommended enabling CloudTrail for audit purposes.
 - **Public Buckets:** I checked S3 buckets for any public access and confirmed that no buckets were publicly accessible except for static website hosting.
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7. Results and Observations

Resource Inventory:

After reviewing all resources, I found one running EC2 instance, six S3 buckets, and nine IAM users. The running EC2 instance was a t3.micro instance, which is the free tier eligible type.

Unused Resources:

I identified several unused resources, including a stopped EC2 instance and old S3 buckets that were not being used. I also found four IAM users with no recent activity.

Excessive Permissions:

I found that the task3-iam-user had AdministratorAccess, which is a high-risk permission. Additionally, the s3-admin-user had full S3 access, which was broader than needed.

Cost Estimate:

I estimated the total monthly cost to be approximately \$8, primarily from the running EC2 instance and S3 storage.

Observations:

The cloud hygiene review helped identify both security and cost issues. Enabling MFA and CloudTrail would improve security, while deleting unused resources and restricting excessive permissions would reduce costs and improve security posture.

8. Possible Improvements for Production

If this were a production environment, I would recommend the following improvements:

- **Regular Resource Reviews:** I would schedule monthly resource reviews to identify and clean up unused resources promptly.
 - **MFA Enforcement:** I would enforce MFA for all IAM users to add an extra layer of security.
 - **CloudTrail Enablement:** I would enable CloudTrail to log all API calls for audit and compliance purposes.
 - **IAM Policy Optimization:** I would review and optimize all IAM policies to ensure they follow the principle of least privilege.
 - **AWS Organizations:** If the environment includes multiple AWS accounts, I would use AWS Organizations for centralized management.
 - **Cost Monitoring:** I would enable cost monitoring and budget alerts to prevent unexpected spending.
 - **Resource Tagging:** I would implement resource tagging to better track resource ownership and purpose.
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9. Cloud Hygiene Report

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Cloud Hygiene Report

Executive Summary

The cloud hygiene review identified several unused resources, excessive permissions, and cost-saving opportunities. The total estimated monthly cost is approximately \$8.

Key Findings

1. **Unused Resources:** A stopped EC2 instance and several old S3 buckets were identified as unused.
2. **Excessive Permissions:** The `task3-iam-user` had `AdministratorAccess`, and `s3-admin-user` had full S3 access.
3. **Missing Security Controls:** MFA was not enabled for any IAM user, and CloudTrail was not enabled.
4. **Cost Savings:** Approximately \$0.58 per month could be saved by deleting unused resources.

Recommendations

1. Remove `AdministratorAccess` from `task3-iam-user`.
2. Restrict `s3-admin-user` to specific S3 buckets.
3. Delete or disable unused IAM users.
4. Enable MFA for all IAM users.
5. Enable CloudTrail for audit logging.
6. Terminate the stopped EC2 instance.
7. Delete unused S3 buckets.

Cost Savings

- Delete unused S3 buckets: ~\$0.50/month
- Terminate stopped EC2: ~\$0.08/month
- ****Total Savings:**** ~\$0.58/month

Conclusion

This task provided a hands-on experience in performing a cloud hygiene review. By systematically inventorying resources, identifying unused resources, reviewing IAM permissions, and estimating costs, I was able to provide actionable recommendations to improve security and reduce costs. The key takeaway is that regular cloud hygiene reviews are essential for maintaining a secure and cost-effective AWS environment.